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GJI Geomagnetism, rock magnetism and palaeomagnetism

High-resolution palaeomagnetic results of Ethiopian trap series from Lima Limo section: implications for the Oligocene geomagnetic field behaviour and timing of volcanism

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SUMMARY

We present high-resolution palaeomagnetic and rock magnetic results from 92 successive volcanic units on a 2 km tall Lima Limo (LL) section of the Oligocene Ethiopian flood basalts (OEFB). A total of 85 well-defined flow-mean characteristic remanent magnetization directions representing seven stable magnetic polarity zones with six transitional/excursional directions were determined. These results indicate the identification of four more polarity reversals than previous palaeomagnetic studies; the resulting magnetostratigraphy has, therefore, more polarity zonation. Comparison of this result with the most recently revised geomagnetic polarity time scale (GPTS, hereafter GPTS2016), by using available radiometric age data, could not find a unique correlation. Instead, it strongly suggests the recording of at least one very brief polarity chron or putative geomagnetic excursion. Our magnetostratigraphic interpretation infers that the relative timing between the OEFB eruptions and the Oi-2 global cooling estimated two possible scenarios: one where the lowermost reverse-polarity (R1) interval for the LL section is correlated to Chron C11r of GPTS2016, same as the interpretations of earlier studies, would make a causal linkage between the OEFB eruptions and the Oi-2 cooling less possible; and, the eruption timing estimated in another scenario correlating the R1 to Chron C12r might preserve the probability of its causal influence in the Oi-2 cooling. The resulting overall mean pole for the LL section of this study, 78.2° N in latitude and 213.0° E in longitude ($A_{95} = 3.4°$, $N = 79$), is closer to other co-eval palaeomagnetic poles from different regions and the reference pole of Africa. Our analysis of geomagnetic palaeosecular variation (PSV) shows that, compared to the Plio-Pleistocene, the Early Oligocene has higher degrees of PSV but its slightly weaker latitude dependence.

Key words: Africa; Magnetostratigraphy; Palaeomagnetic secular variation; Palaeomagnetism; Large igneous provinces; Volcano/climate interactions.

1 INTRODUCTION

The Oligocene Ethiopian flood basalts (OEFB; Fig. 1a), one of the youngest large igneous provinces in the world, cover a surface area of ~600 000 km^2 (and not less than 750 000 km^2 when considering the part removed by erosion) with an estimated erupted volume of ~350 000 km^3 (e.g. Mohr & Zanettin 1988). In Lima Limo area ('LL' in Fig. 1a) its total thickness reaches up to ~2 km (e.g. Hofmann *et al.* 1997). The OEFB province extends on the opposite part of the Red Sea into Yemen (e.g. Coulié *et al.* 2003; Riisager *et al.* 2005). This extensive flood volcanism is generally linked to the breakup of the Afro-Arabian plate (Abbate *et al.* 2015). The large volume of the volcanism, coupled with its tall sequence, has attracted a great deal of scientific research interest to investigate the

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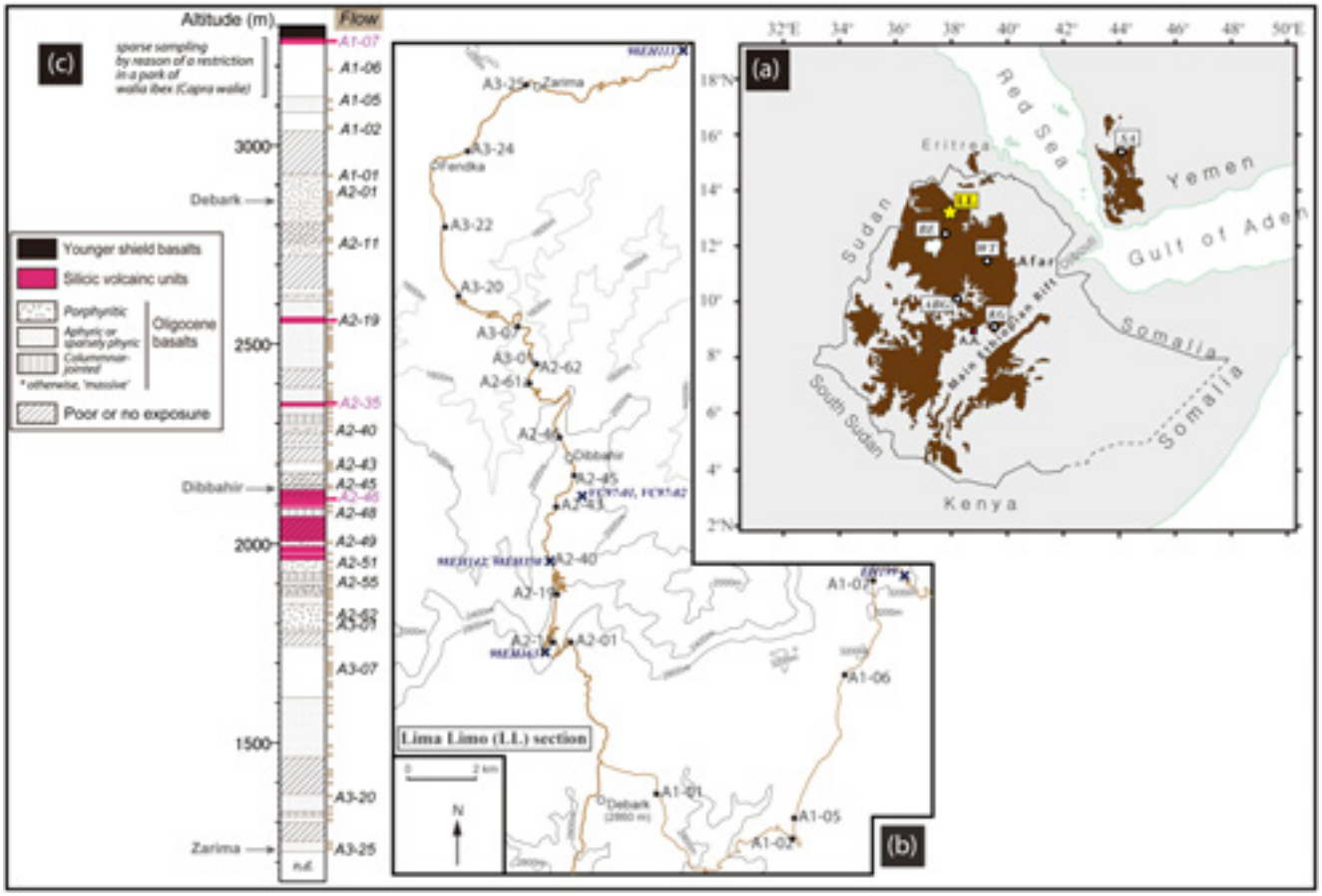


Figure 1. Map of the Oligocene Ethiopian-Yemen traps (i.e. OEFB), and locations of the LL section sampled in this study and stratigraphic log of the volcanic units therein. (a) Generalized map showing the distribution of OEFB traps and the location of the LL section (yellow star), with some locations of previous studies for comparison (ABG: Abbay Gorge, BE: Belesaa, KG: Kessem Gorge, SA: Sana'a, WT: Wegel Tena,). A.A. (red square): Addis Ababa. (b) Detailed location map of the sampling sites. Note that not all sites are marked. Some locations where available radiometric ages were reported are also shown as 'X' symbol with their original identification codes. (c) Stratigraphic log of the LL section during the 2012 sampling campaign.

Oligocene geomagnetic field behaviour and to assess the possible linkage between OEFB volcanism and global climate change (e.g. Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003; Peate *et al.* 2003; Touchard *et al.* 2003; Riisager *et al.* 2005; Lhuillier & Gilder 2019).

Hofmann *et al.* (1997) and Rochette *et al.* (1998) conducted pioneering magnetostratigraphic, K-Ar and $^{40}Ar/^{39}Ar$ investigations on LL and Wegel Tena ('WT' in Fig. 1a) sections of the OEFB and made a general interpretation that the major part of the OEFB erupted around 30 Ma with a short period of ∼1.5 Myr or less. These authors identified R-N-R polarity sequence and eventually correlated it to magnetochrons C11r-to-C11n or C11r-to-C10r in the geomagnetic polarity time scale (GPTS; Fig. 2). Coulié *et al.* (2003) had undertaken K/Ar and $^{40}Ar/^{39}Ar$ geochronologic analyses and improved the correlation of the Rochette *et al.*'s magnetostratigraphy with the GPTS by Cande & Kent (1995, hereafter termed CK95) (see Fig. 2b). Lhuillier & Gilder (2019) conducted palaeomagnetic analyses with K-Ar geochronology on a ∼1 km tall section in Belessa area ('BE' in Fig. 1a), South Gondar, where shield volcanoes and associated flood basalts, as young as ∼23 Ma (Kieffer *et al.* 2004), are reportedly present. They revealed four stable polarity intervals and the lowermost horizon with normal polarity, suggesting correlation to chrons C11r-to-C11n.1n in the GPTS by Ogg (2012) and the lowermost normal-polarity record possibly as the end of Chron C12n or excursional in Chron C11r. Despite these efforts, the current magnetotostratigraphy and radiometric chronology data have yet to provide a unique and robust solution in magnetostratigraphic correlation for the OEFB.

Hofmann *et al.* (1997) and Rochette *et al.* (1998) have also suggested the onset of OEFB-related basaltic and silicic volcanism to be during C11r and C11n, respectively, and thus implying a possible temporal linkage between the OEFB volcanism and the Oi-2 global cooling event, which was represented by a maximum in the oxygen isotope record and dated at ∼30.3 Ma by Pekar & Miller (1996) or 30.35 Ma by Wade & Pälike (2004). This possible linkage was later discussed by several studies (Peate *et al.* 2003; Touchard *et al.* 2003; Riisager *et al.* 2005). Touchard *et al.* (2003) correlated four rhyolitic tephra layers found in the ODP core 711A (Indian Ocean) to the OEFB-related silicic volcanism with aerial emissions and advocated the earlier suggestion. On the other hand, Peate *et al.* (2003) correlated the two tephra layers sampled by the ODP core 711A to the OEFB-related on-land ignimbrites in Yemen and suggested that the OEFB silicic volcanism postdates the Oi-2 event, documenting a virtually impossible causal link between them but leaving room for possible exacerbation of climatic change by such large-volume silicic volcanism. In these two papers, there is a clear difference in the age constraints of the tephra layers: Touchard *et al.* (2003) was based on the correlation of their core 711A bio-magnetostratigraphy

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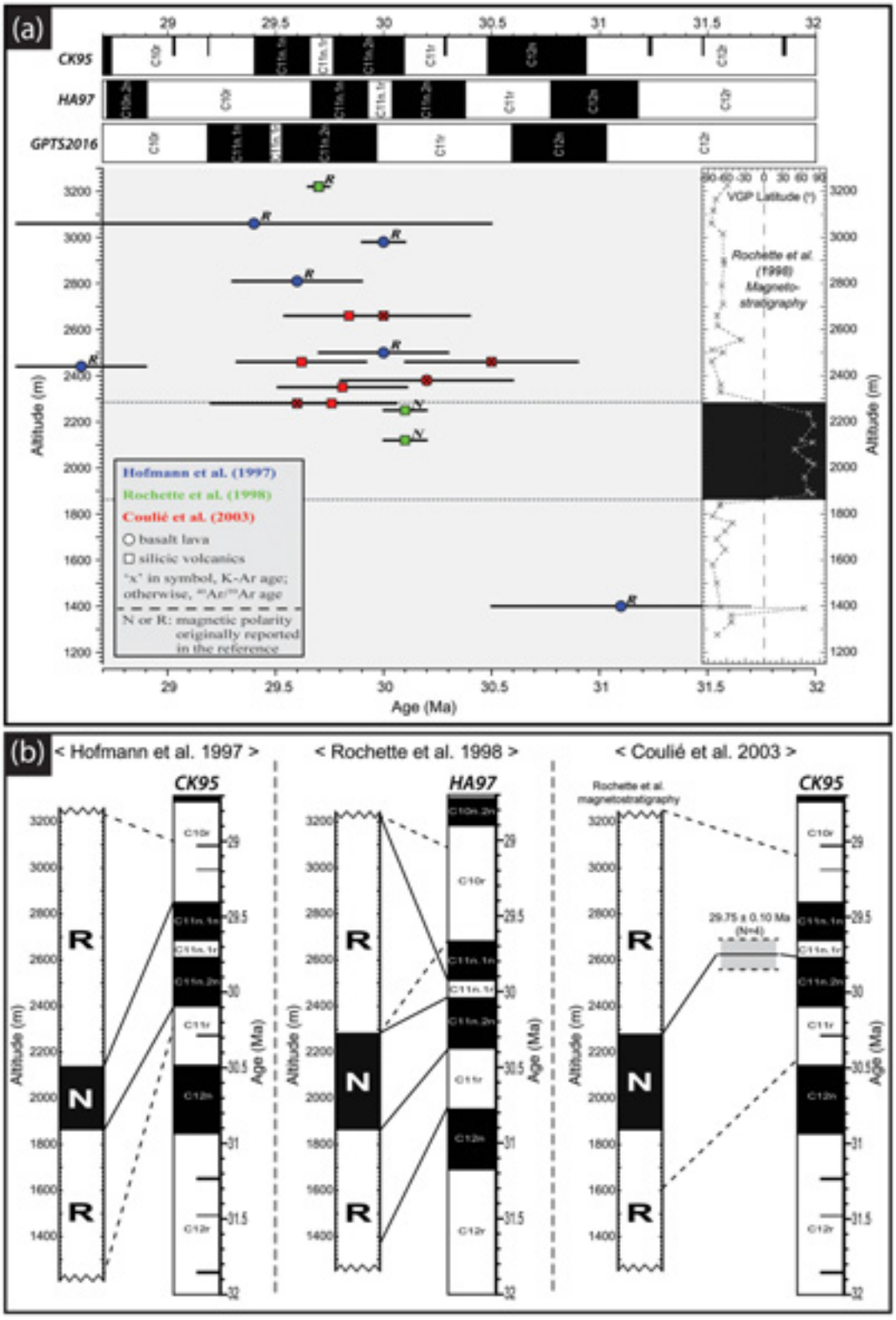


Figure 2. Summary of previous radiometric age data, magnetostratigraphy and GPTS correlations for the LL section. (a) Selected $^{40}Ar/^{39}Ar$ and K-Ar age estimates with 1σ uncertainties from Hofmann *et al.* (1997), Rochette *et al.* (1998) and Coulié *et al.* (2003) are plotted on an altitude versus age diagram. On the top of the altitude–age plot, the different versions of the reference global GPTS are illustrated: CK95 (Cande & Kent 1995; Cryptochrons therein reported are also shown as black half-bars), HA97 (Huestis & Acton 1997) and GPTS2016 (Ogg *et al.* 2016). On the right of the altitude–age plot, the VGP latitude data with altitudes of the LL section reported by Rochette *et al.* (1998) are shown, with their resulting magnetostratigraphy (black: normal polarity and white: reversed polarity). (b) Previous magnetostratigraphic interpretations (Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003) correlating the LL magnetostratigraphic result (the work of Hofmann *et al.* 1997 or Rochette *et al.* 1998) to a reference GPTS (the work of Cande & Kent 1995 or Huestis & Acton 1997).

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to the integrated timescale by Berggren *et al.* (1995), which calibrated and integrated the calcareous nannofossil zonation to CK95; Peate *et al.* (2003) was based on their geochemical and isotopic correlations to the on-land silicic ignimbrites in Ethiopia and Yemen having available $^{40}Ar/^{39}Ar$ ages. It is good to note that the Touchard *et al.*'s bio-magnetostratigraphic age control would include nontrivial uncertainty because of the significant revision of the NP23/NP24 boundary in calcareous nanofossil zonation in the latest Palaeogene timescale (Ogg *et al.* 2016). Riisager *et al.* (2005) employed palaeomagnetic and $^{40}Ar/^{39}Ar$ investigations from Yemen sections of the OEFB, and documented that the basal basaltic volcanism in Yemen occurred apparently during C11r thus suggesting that a causal role of the onset of OEFB basaltic volcanism in the Oi-2 event cannot be ruled out. These previous constraints on the relative timing between the two events are thus insufficient yet in reaching a clear consensus on the possibility of a causal link between them.

It is obvious that the current determinations of the respective ages for the OEFB and Oi-2 events and the relative timing between them rely on the resolution of magnetostratigraphic constraints used and determined. Several factors contribute to the related ambiguity; for example, the majority of the uncertainties of previous age estimates for the OEFB (Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003; see Fig. 2a and Table S1, Supporting Information) are comparable with the durations of magnetochrons for the corresponding geologic period (<0.1 Myr at minimum); the previous studies that documented the possible link between OEFB and Oi-2 events used CK95 or the GPTS by Huestis & Acton (1997; hereafter termed HA97), where there is age discrepancy of up to ∼0.3 Myr between CK95 and HA97 for the Early Oligocene. Now the latest GPTS by Ogg *et al.* (2016), which has significant differences in age for the respective Chron boundaries from the earlier GPTSs (Fig. 2a), is available.

Several palaeomagnetic studies of OEFB also investigated geomagnetic palaeosecular variation (PSV) of the geomagnetic field and palaeomagnetic pole position during the Oligocene. Rochette *et al.* (1998) investigated the LL (sampled sites, $N = 42$) and WT (sampled sites, $N = 23$) sections in Ethiopia, and found the angular standard deviation (ASD) value as PSV estimation of ∼15°, and an overall mean pole far-sided by ∼6° with respect to the synthetic reference pole of Africa, suggesting a sizeable non-dipolar component of the Oligocene field (∼15 per cent quadrupole component). On the other hand, Riisager *et al.* (2005) yielded a palaeomagnetic pole of latitude (Lat.) = 74.2° N, longitude (Lon.) = 249.1° E ($A_{95} = 3.6°$, $N = 48$) and ASD of 14.2° from coeval Yemeni sections, and suggested prevailing axial dipole field with insignificant non-dipolar contributions in the Oligocene field based on their global analysis of coeval palaeomagnetic poles. Recently, Lhuillier & Gilder (2019) added palaeomagnetic data from the Belessa section (sampled sites, $N = 63$) and analysed PSV and elongation of data distribution by using the data set combining both Ethiopian and Arabian OEFB sections ($N = 167$). They thus suggested higher ASD during the Early Oligocene than during the Plio-Pleistocene and the axial dipole-dominated field during the past 30 Myr. However, we know that a larger number of time-independent units analysed for a single section is favourable in robust PSV estimation.

The primary objective of this work is to revisit the LL section (∼2 km tall) and employ field techniques identifying more time-independent cooling units and conduct palaeomagnetic investigation on a larger number of lava flows than the earlier studies, thereby refining its magnetostratigraphy, palaeomagnetic pole and PSV, and further helping refinement of the relative timing between OEFB and Oi-2 event.

2 GEOLOGICAL BACKGROUND AND SAMPLING

The LL section for this study is located ∼52 km to the west of the peak Ras Dashen of the Simien Mountains. It extends laterally from the town of Zarima, in the north (which is at the base of the flood basalts and top of Mesozoic-aged sandstone unit), to Debark, near the transition from the plateau to the flank of the Simien volcano, to the south (Fig. 1b). It consists of mainly flood basalts with tholeiitic to transitional affinity, which are overlain by massive Simien shield basalts with tholeiitic compositions (∼30 to ∼19 Ma) and other numerous but volumetrically small shield volcanics mainly composed of alkaline lavas (∼23 to ∼ 11 Ma, Kieffer *et al.* 2004). The basaltic sequence is intercalated with some silicic units (see Fig. 1c). A detailed description of the section including lithological, petrological and chemical data are given in Kieffer *et al.* (2004). According to Kieffer *et al.* (2004), the overlying Simien basalt series are lying above ∼2670 m altitude and laterally over an area of the town of Debark. They also observed that the Simien series were in contact with the flood basalts having small dips of 3°–4° away from the summit of the Simien volcano, but the flood basalts were generally flat-lying. In the uppermost part of the LL section, there was a thin (∼10 m) silicic unit overlying the Simien basalts.

Our sampling was undertaken along the roads between the town of Zarima town and near the Simien Lodge which is located ∼10 km to the east of the town of Debark (Fig. 1b) in 2 weeks in 2012 March–April. It covered the OEFB sequence of ∼1920 m thick (Fig. 1c). The sampled rocks were from basaltic lava flows and silicic intercalations located at 1960–2140, ∼2354, ∼2560 and ∼3250 m altitudes. Stratigraphic levels of sampled units were recognized by field observation and altitude. We also observed that the lower section below ∼2200 m was marked by thin lava flows (<10m) and a gentle slope morphology. In contrast, the upper part (above ∼2200 m) was constituted mainly of thick lava flows (10–100 m) generating a cliff with a clear break along with flow contacts. Several large-scale terraces were seen, but neither thick palaeosols nor interlayered sediments were found. Owing to poor exposures at ∼2000–2070, ∼2640–2740 and ∼2900–3040 m horizons and entrance restrictions at 3120–3260 m levels, sampling densities were sparse (Fig. 1c).

A total of 97 palaeomagnetic sites were sampled: 93 basalts and four silicic units. A palaeomagnetic site corresponds to one flow or cooling unit. The sites were labelled as A1-xx (elevations above the town of Debark), A2-xx (below the town of Debark down to ∼1800 m) or A3-xx (below ∼1800 m) (Figs 1b and c). All sites (except for sites A1–02 to A1–07) were directly within a vertical sequence running roughly N-S direction for ∼20 km distance. Sites A1–02 to A1–07 were sampled along a roughly E-W direction further to the east from the town of Debark. Stratigraphic height between sites varied from ∼1 to ∼120 m, depending on the thickness of individual flows/units or, sometimes, outcrop exposure, with an average of ∼20 m. A total of 399 oriented blocks with ∼10–15 cm diameters were collected (on average four blocks per site) from the interior part of a flow/unit. Core axis orientating for palaeomagnetic analyses was made using a Brunton magnetic compass. Any influence by activities such as a bolt of lightning strike for the majority of the sample was examined by means of a magnetic compass, which was placed several meters away from the sampling spot to see if there was any deflection of the compass needle. The maximum horizontal distance between block samples within site varied from a few metres to ∼20 m with sites. The present-day geomagnetic field declination at the sampling locality is 2.6° (Finlay *et al.* 2010), a

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value which is corrected for all palaeomagnetic analyses. Localities and altitudes of sampling sites were determined by using handheld GPS equipment. The altitude readings were made by GPS, which were double-checked and corrected by a barometric altimeter at some sites where overhanging blocks jammed GPS signals. From visual inspections of the outcrops and the horizontal continuity of exposed cliffs over more than ∼10 km, no bedding correction of sampled horizons appeared to be necessary as they are horizontally emplaced.

3 METHODS

All laboratory and experimental procedures were performed at the palaeomagnetic laboratories either at Kobe University or at the Center for Advanced Marine Core Research (CMCR), Kochi University, in Japan.

Several cylindrical cores with 25 mm diameter were drilled into each oriented block samples, and generally, each drilled core was sliced into multiple specimens of ∼20 mm height for palaeomagnetic analyses (the cut specimen is termed STD specimen). Some of the cut cores were further cut and divided into smaller specimens with sizes of 3/5th, half, 2/5th, 1/5th and 1/10th STDs, for the purpose of additional rock-magnetic analyses and due to strong magnetic moments above a sensitivity of the magnetometers used: the shape of these small specimens is cylindrical or cubic where the volume of each specimen was assumed to be 10, 6, 5, 4, 2 and 1 cm^3, respectively, in calculations of magnetization intensity and magnetic susceptibility.

The specimens, after measurements of low-field magnetic susceptibility at room temperature using a Bartington Instruments MS2B meter, were subjected to a progressive alternating field (AF) or thermal (TH) demagnetization to isolate a characteristic remanent magnetization (ChRM) component. Two types of progressive AF demagnetizations (PAFD) experiments were carried out: only a PAFD technique, and a hybrid technique of PAFD with a low-temperature demagnetization (LTD). The LTD treatment was to soak the specimens into liquid nitrogen within a cylindrical mu-metal magnetic shield for 10 min prior to PAFD. For only pilot specimens (one specimen per flow), the former employed 15–17 AF steps up to a maximum of 100 mT using a DEM-93 AF demagnetizer (Natsuhara Giken), whereas the latter employed 38 steps with step intervals of 2 mT (up to 30 mT), 5 mT (30–100 mT) and 10 mT (100–180 mT) up to a maximum AF of 180 mT, using an automated AF demagnetizer equipped with a DSPIN-2 spinner magnetometer (Natsuhara Giken). For TH demagnetization, 14–16 progressive heating steps up to 580–620 °C (with intervals of 50 °C for 100–400 °C, 30 °C for 400–520 °C and 20 °C for above 520 °C) were employed using two TDE-91C TH demagnetizers (Natsuhara Giken). Natural remanent magnetizations (NRMs) were measured with an SMM-85 spinner magnetometer (Natsuhara Giken) except for the hybrid LTD-PAFD experiments.

Demagnetization results were identified using orthogonal vector component diagrams and equal-area projections. To isolate and define the ChRM component leading to determination of palaeomagnetic direction, principal component analysis (PCA) using the best-fit line of Kirschvink (1980) or remagnetization great circle (GC) of Halls (1976) were used. A set of criteria was that a site should have three or more data points and that the maximum angular deviation (MAD) value, of the anchored line and GC fitting, should be below 12°. Mean directions were usually calculated using the Fisherian statistics (Fisher 1953), whereas statistics by McFadden & McElhinny (1988) and co-authors were used when including the GCs. All of these data analyses were made by using the PaleoMac software package (Cogné 2003).

Various rock magnetic properties were measured on tens of grams of small fragments to characterize ferrimagnetic minerals of the remanence carrier. The measurements included magnetic hysteresis loops, backfield curves and progressive acquisitions of isothermal remanent magnetization (IRM) with a MicroMag 3900 vibrating sample magnetometer (Princeton Measurements Corporation), and strong-field thermomagnetic analyses with an NMB-89 magnetic balance (Natsuhara Giken). The hysteresis parameters are thought to give information on granulometry of the magnetic minerals by using the Day diagram (Day *et al.* 1977) together with the theoretical lines of particle mixing by Dunlop (2002). Each saturation magnetizations versus temperature (*Ms–T*) curve through thermomagnetic analyses was conducted by heating specimens to 700 °C and cooling them to room temperature in a constant field (0.3 or 0.5 T) under vacuum (1–10 Pa). Determination of the Curie temperature (*Tc*) was made by the intersecting tangents method (Prévot *et al.* 1983).

4 RESULTS

Palaeomagnetic and rock magnetic measurements were carried out on 92 sites (91 basaltic flows and one silicic unit); samples of three silicic (A1–07, A2–35 and A2–46) and two basaltic sites (A2–47 and A2–50) were not used owing to their relatively high friability (indicating poor welding of the silicic unit or severe alteration of the basalt samples). 562 specimens were subjected to the progressive demagnetization–NRM measurement procedures. One specimen (sometimes, two) per flow were used in the set of rock magnetic measurements.

4.1 Rock magnetic properties

Site averages in NRM intensity and magnetic susceptibility showing in range of 0.26–39.1 A m^{-1} and 4.0–89.1 × 10^{-3} SI, respectively (Table S2, Supporting Information) are comparable to those reported by Rochette *et al.* (1998). A part of samples in the sites with > 8 A m^{-1} in NRM intensity were suspected of having the overprint of IRMs induced by lightning strikes; hence their demagnetization results were carefully interpreted in ChRM determination (see Section 4.2).

Ms–T curves reveal the presence of one or two ferrimagnetic phases whose *Tc* is variable, ranging mostly between ∼190 and ∼590 °C (Fig. S1 and Table S2, Supporting Information), and the IRM acquisition experiments show saturation by ∼80–300 mT (Fig. S2b, Supporting Information) indicative of little or no high-coercivity phases. It is thus interpreted that the majority of the samples contain one phase of Ti-poor titanomagnetite or titanomagnetite with significant Ti content, and some contain both. In a few specimens, presence of (titano-)maghemite is detected by a substantial decrease in magnetization intensity above 300 °C followed by a hump peak of the intensity between 400 and 500 °C (e.g. Fig. S1k, Supporting Information). Magnetic hysteresis parameters generally fall close to the SD–MD mixing curves outlined by Dunlop (2002) although a few falls towards the SP mixing line (Fig. S2, Supporting Information).

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4.2 Palaeomagnetic directions

Demagnetization behaviours for specimens from each sample with different demagnetization treatments (AF, LTD–AF and TH) generally allow to recognize remanence components with identical direction between one another. The typical demagnetization results obtained are shown in Fig. 3. For the majority of analysed specimens, the low coercivity/low unblocking temperature component of remanence was removed by 5–20 mT in AF demagnetization or by a temperature of 250–300 °C in TH demagnetization. After these steps, a high-stability component, which is directed towards the origin and defining a straight line, was isolated. This component is interpreted as ChRM acquired during original cooling (Figs 3a and b). The NRM directions isolated at temperature above *Tc* of pure magnetite are not significantly different from the ChRM direction isolated at lower temperatures for most specimens. On the other hand, the suspected lightning-struck specimens exhibit similar behaviour as shown in Fig. 3(c), where the low- and high-stability components are not preferentially isolated due to significant overlap in their coercivity and unblocking temperature spectrum. Generally, AF technique seems to be more efficient than TH one in removing the low-stability components. For some cases with overlapping components, a ChRM vector is determined by the GC fit (see Fig. 3c). As a result, we determine ChRM direction vectors from 554 specimens: 514 PCA-derived linear directions and 40 GC fits (Table S3, Supporting Information).

From each of the 92 sites, at least four ChRM vectors are determined, and they generally have good within-site consistency. However, within site in some cases, there exists an outlier ChRM vector or vectors deviated conspicuously from most other ChRM vectors having a similar direction. These outlier directions are considered to originate from either personal error during orienting in field sampling, or specimen(s) laboratory preparation, or due to incomplete removal of the overprint remanence. After screening out these outliers, collectively 538 ChRMs (an average of approximately 6 per flow) is used for calculations of flow-mean ChRM direction (Table S3, Supporting Information).

The resulting flow-mean ChRM directions are listed in Table 1. For all sites except for four sites (A1–01, A2–42a, A3–07 and A3–23), a mean ChRM direction could be determined per flow. The majority of the sites reveal well-defined mean directions with 95 per cent confidence level (α_{95}) less than 8° and precision parameter (k) values between 43 and 982. Three sites A2–10, A2–12 and A2–17, however, give poorly defined directions: $\alpha_{95} > 12°$ and $k < 30$. These large dispersions are thought to be caused by incomplete ChRM isolation from the secondary components. From sites A1–01, A2–42a, A3–07 and A3–23, two distinct clusters in ChRM direction within single flow were identified. In the flows A2–42a, A3–07 and A3–23, the identified two clusters are spaced less than 1 m apart in stratigraphic position and hence the two clusters might result from remagnetization by overlying lava flows. Within the flow A1–01, likely, one ChRM mean (declination (D) = 337° and inclination (I) = 37°) may correspond to a secondary direction remagnetized possibly by a lightning strike and the other (D = 252° and I = 19°) likely to be a primary direction, but both are poorly defined ($k < 20$ and $\alpha_{95} > 14°$). Six flow means of A2–42a_up, A2–42a_lo, A2–53, A3–07b, A3–23_up and A3–23_lo are from two block samples so that it is hard to consider as high-reliability primary directions. Hence, these 11 mean directions are excluded from further considerations.

As a result, we obtain 85 well-defined flow-mean ChRM directions and the corresponding virtual geomagnetic poles (VGPs, that are calculated by using individual site latitudes and longitudes), as given in Fig. 4 and Table 1. Distribution of the flow-mean directions is clustered into two opposite polarity-magnetized directions, despite some scattered sites and pass the reversal test (McFadden & McElhinny 1990) with the classification 'C' at 95 per cent confidence level for both cases including and excluding the scattered data (see also Section 4.4). This data set also passes the Cartesian bootstrap reversal test (outlined in Tauxe 2010), which were computed using 'Paleomagnetism.org' (Koymans *et al.* 2016), an online toolset for analysing palaeomagentic data.

Fig. 4(b) shows that the distribution of the ChRM directions is elongated on the meridional plane, whereas the VGP distribution is more circular, which is consistent with that predicted by the TK03.GAD statistical model for the past 5 Myr (Tauxe & Kent 2004).

4.3 Magnetostratigraphy

Variations of ChRM declinations and inclinations and VGPs in stratigraphic order are shown in Fig. 4(a). The LL magnetostratigraphy is interpreted to contain seven magnetic polarity zones, namely R1−N1−R2−N2−R3−N3−R4 from bottom to top, separated by six complete polarity reversals. Compared with the previous simple R-N-R sequence (Hofmann *et al.* 1997; Rochette *et al.* 1998), our results have four more geomagnetic reversals and higher thickness ratio of normal to reversed polarities (Fig. 4a). Five (out of six) reversals are situated at a narrow stratigraphic interval around the horizon where the previously reported R-to-N reversal was recognized. The thicknesses for magnetozones N1, R2, N2 and R3 are remarkably smaller than those for the others. For the magnetic polarity intervals that are newly presented in this study, that is, N1 to R3, the NRM demagnetization behaviours and thermomagnetic curves (Fig. S3, Supporting Information) indicate, no doubt, that they preserved the primary palaeomagnetic records.

Apparently, excursional records are also found in R1 and R4 (see Section 4.4). Silicic horizons are in magnetozones N3 and R4.

When correlating our interpreted magnetostratigraphy to the latest GPTS (Ogg *et al.* 2016; termed hereafter GPTS2016) in conjunction with the available radiometric ages (Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003), there is no unique correlation of the individual magnetozones to stable polarity chrons in GPTS2016. Discussion on this will be given in Section 5.1.

4.4 Palaeomagnetic pole position and palaeosecular variation

The determined VGPs are concentrated on an area of latitudes higher than 60° N, but some VGPs are far removed from the cluster (Fig. 4b). Palaeomagnetic pole position and secular variation as a time-averaged characteristic of the geomagnetic field are determined by using site mean (i.e. flow-mean) VGP data of stable polarity state (i.e. not excursional/transitional). We use the cut-off method of Vandamme (1994) to determine the optimal cut-off angle in defining a distribution of 'stable polarity' VGP data. From this, the optimal cut-off angle is determined 35.9° (Table 2). Thus six data from sites A1–02, A2–11, A2–20, A2–21, A2–40 and A3–20 are estimated to be excursional/transitional: A2–40 indicates transitional during the N3-to-R4 reversal, while the others are excursional. The

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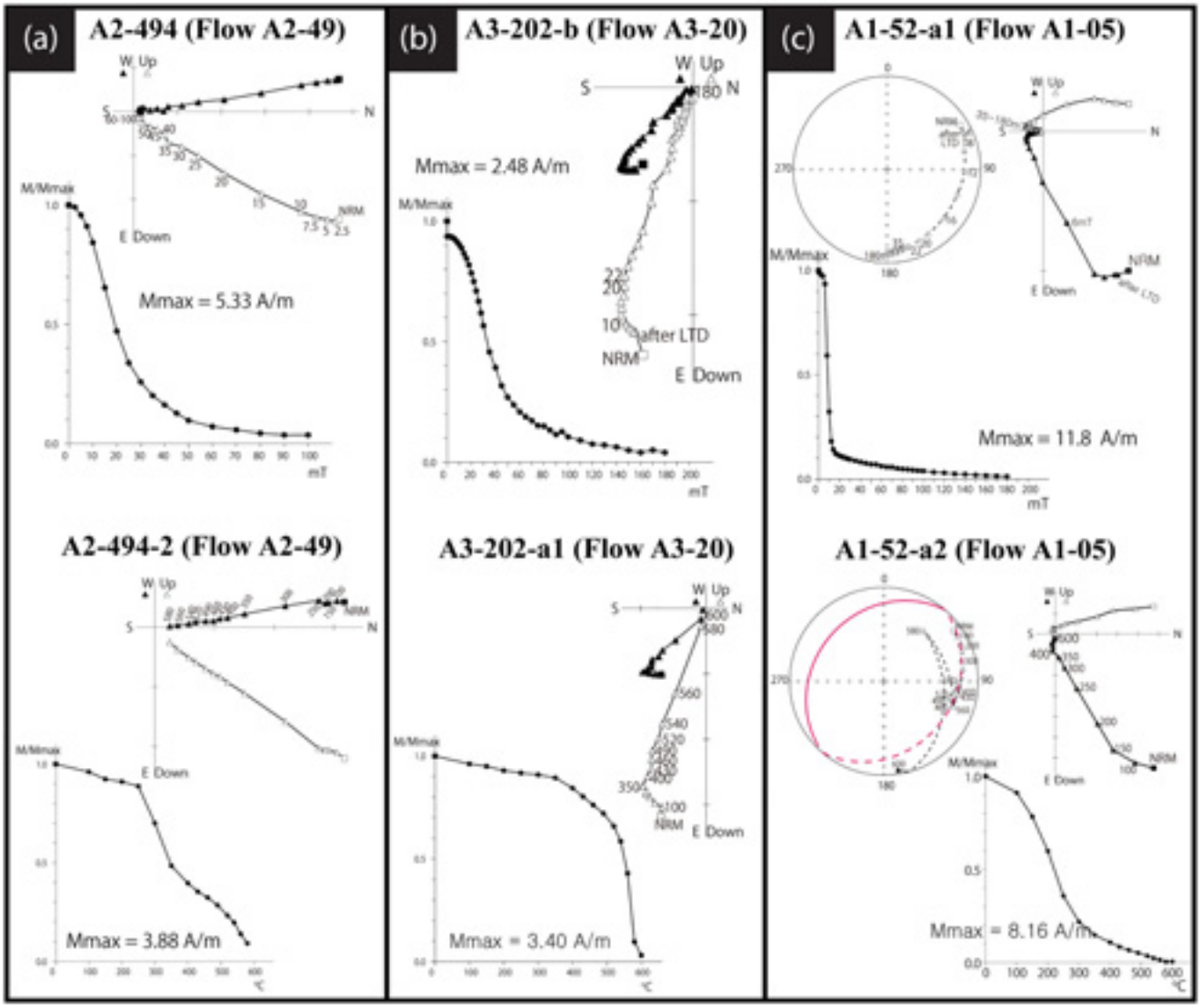


Figure 3. Typical progressive demagnetization results for three pairs of specimens. (a) The respectively AF (upper part) and TH (lower part) subjected specimens for flow A2–49, (b) for flow A3–20 and (c) for flow A1–05. For each demagnetization result, both of orthogonal vector diagram and NRM intensity decay curve are shown, and additionally in Fig. 3(c) stereographic projection is also shown. In the orthogonal vector diagram solid (open) symbol shows projection onto the horizontal (vertical) plane. In the stereographic projection solid (open) symbol shows downward (upward) inclination, and a pink curved line indicates the plane of a GC fitting using data of TH steps from room temperature to 560 °C. See the text for details.

six data are excluded in palaeomagnetic pole and PSV determination. The remaining 79 VGPs are used to obtain a mean pole for LL as a palaeomagnetic pole at ∼30 Ma, resulting in Lat. = 78.2° N and Lon. = 213.0° E ($A_{95} = 3.4°$; Table 2).

PSV is estimated as ASD of the VGP distribution. If Δ_i is the angular distance of the *i*th VGP from a reference pole (here we used the calculated mean pole from our determined VGPs), the ASD (S_P) of N VGPs is given by $S_P = (\frac{1}{N-1}\sum_{i=1}^{N}\Delta_i^2)^{1/2}$ (Cox 1970). But S_P is a composite of the ASD as between-site dispersion (S_F) that arises from the geomagnetic field and the within-site dispersion (S_W) that occurs from experimental and sampling errors so that the observed S_P must be corrected for the contribution of S_W. The ASD we seek (S_F), therefore, is given by $S_F^2 = S_P^2 - \frac{1}{N-1}\sum_{i=1}^{N}(S_{W_i}^2/n_i)$ (e.g. Biggin *et al.* 2008), where S_W is the within-site dispersion associated with each VGP, calculated from the precision parameter of the within-site distribution in pole space (K), and n is the number of samples within the site (in practice, we here used the number of specimens in calculation). We note that, within each site, most of the specimens that provide individual VGP data are from different cores while, in a few cases, one or two pairs of specimens are from single cores (Table S3, Supporting Information). We here consider that each core can constitute a sample and that this consideration of (near-)core-level data within site does not make systematic orientation errors. The pole-space precision parameter, K, is translated from the direction-space precision parameter, k, with the reasonable assumption of a Fisherian distribution (Fisher 1953) of within-site directions, by the following formula: $K = 8\ k/(5 + 18\sin^2\lambda + 9\sin^4\lambda)$ (Cox 1970), where λ is the palaeolatitude of the sampling site, calculated from the determined site-mean inclination, I, by the dipole relation: $\lambda = \tan^{-1}(\tan I/2)$.

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Table 1. Summary of palaeomagnetic field directions and the corresponding VGPs for the LL section (Ethiopia). Flow, label given at the flow sampled in the field; DG, directional group (see the text for details); Altitude, altitude reading of the flow from handheld GPS with an altimeter to crosscheck; Latitude (Longitude) in Locality, latitude (longitude) of each sampling location; n_0, number of specimens measured from the flow; n, number of specimens used to determine a flow-mean direction; n_l (n_{gc}), number of specimens with ChRM directions determined by a linear fit (GC fit); D (I), mean ChRM declination (inclination) for the flow; k, Fisherian precision parameter of the mean direction; α_{95}, 95 per cent confidence limit on the mean direction; Latitude (Longitude) in VGP, latitude (longitude) of the VGP; Pol., geomagnetic polarity determined by the geographic position of VGP (N: normal polarity, R: reversed polarity, -i: intermediate northerly or southerly position). *n.a.* indicates 'not analysed' data in further discussion of this study.

			Locality										VGP		
Flow	DG	Altitude (m)	Latitude (° N)	Longitude (° E)	n_0	n	n_l	n_{gc}	D (°)	I (°)	k	α_{95} (°)	Latitude (° N)	Longitude (° E)	Pol.
A1–07		3261	13.2094	37.9729											
A1–06		3190	13.1843	37.9653	9	9	3	6	188.3	− 54.3	137.6	4.4	− 67.1	235.7	R
A1–05		3112	13.1464	37.9514	7	7	5	2	177.1	− 13.4	246.4	3.9	− 83.0	62.4	R
A1–04		3089	13.1422	37.9511	7	7	7	0	173.9	20.0	202.7	4.3	− 65.8	52.7	R
A1–03		3047	13.1408	37.9509	7	7	7	0	197.4	20.1	55.1	8.2	− 60.8	0.8	R
A1–02		3041	13.1412	37.9504	7	7	7	0	225.8	33.7	124.6	5.4	− 34.9	341.9	R-i
A1–01		2923			10										
			13.1527	37.9139		5	4	1	252.3	18.8	14.0	14.0			*n.a.*
			13.1527	37.9139		5	5	0	337.1	37.3	19.0	18.0			*n.a.*
A2–01		2880	13.1931	37.8908	7	6	6	0	174.5	24.8	263.0	4.1	− 63.2	49.9	R
A2–02		2869	13.1919	37.8899	8	7	7	0	173.2	15.9	400.0	3.0	− 67.7	55.8	R
A2–03		2859	13.1918	37.8896	8	8	8	0	172.4	10.8	223.1	3.7	− 69.9	60.4	R
A2–04		2854	13.1926	37.8899	8	8	8	0	170.3	8.4	147.3	4.6	− 70.1	67.5	R
A2–05		2850	13.1934	37.8900	7	6	6	0	176.0	8.0	269.6	4.1	− 72.3	51.1	R
A2–06		2822	13.1909	37.8884	7	7	7	0	176.2	11.4	83.2	6.7	− 70.7	49.4	R
A2–07		2817	13.1909	37.8884	6	5	4	1	175.6	8.2	381.3	3.9	− 72.1	52.3	R
A2–08		2815	13.1906	37.8881	6	4	4	0	172.7	7.4	251.4	5.8	− 71.6	61.6	R
A2–09		2809	13.1895	37.8873	6	6	6	0	180.4	7.3	96.4	6.9	− 73.1	36.5	R
A2–10		2766	13.1939	37.8867	6	5	5	0	187.6	17.9	26.1	15.2			*n.a.*
A2–11		2756	13.1931	37.8860	6	5	1	4	111.4	− 21.2	145.8	6.4	− 23.1	134.4	R-i
A2–12		2727	13.1958	37.8861	6	5	3	2	222.2	− 13.1	14.6	20.7	− 48.0	312.6	*n.a.*
A2–13		2634	13.1921	37.8837	5	5	5	0	182.3	− 1.0	603.8	3.1	− 77.1	27.5	R
A2–14		2628	13.1929	37.8841	6	6	6	0	150.1	1.7	981.8	2.1	− 57.2	104.8	R
A2–15		2623	13.1939	37.8842	5	5	5	0	185.4	20.7	296.9	4.4	− 65.5	25.0	R
A2–16		2603	13.1967	37.8851	5	4	4	0	177.9	5.2	156.9	7.4	− 74.1	45.5	R
A2–17		2589	13.1985	37.8864	6	6	6	0	171.2	− 29.6	8.7	24.1			*n.a.*
A2–18		2578	13.2000	37.8862	6	6	6	0	172.3	− 10.5	168.6	5.2	− 79.0	82.4	R
A2–19		2559	13.2058	37.8871	6	3	3	0	180.3	3.3	168.9	9.5	− 75.1	36.7	R
A2–20		2544	13.2069	37.8893	6	6	6	0	215.0	40.5	450.6	3.2	− 40.1	354.3	R-i
A2–21		2537	13.2071	37.8889	6	6	6	0	214.9	40.9	403.2	3.3	− 39.9	354.7	R-i
A2–22		2520	13.2073	37.8886	5	5	5	0	183.8	− 15.7	582.2	3.2	− 83.6	1.9	R
A2–23		2515	13.2074	37.8877	5	5	5	0	178.3	− 36.8	190.3	5.6	− 82.5	205.6	R
A2–24		2506	13.2074	37.8869	4	4	4	0	175.4	− 33.6	148.2	7.6	− 83.2	177.9	R
A2–25		2500	13.2075	37.8869	6	6	6	0	176.2	− 36.4	477.7	3.1	− 82.1	191.0	R
A2–26		2493	13.2078	37.8888	4	4	4	0	188.9	19.1	568.0	3.9	− 65.3	16.5	R
A2–27		2483	13.2076	37.8896	5	5	5	0	190.1	17.8	454.9	3.6	− 65.5	13.2	R
A2–28		2450	13.2077	37.8897	5	5	5	0	190.4	16.8	157.1	6.1	− 65.9	12.0	R
A2–29		2442	13.2079	37.8890	7	7	7	0	189.6	18.4	164.4	4.7	− 65.4	14.6	R
A2–30		2440	13.2085	37.8886	5	5	4	1	176.8	− 15.6	295.4	4.5	− 83.9	69.1	R
A2–31		2399	13.2116	37.8862	5	5	5	0	190.8	− 13.8	386.0	3.9	− 77.7	337.1	R
A2–32a		2382	13.2112	37.8860	5	5	4	1	196.1	34.4	220.2	5.2	− 54.2	11.2	R
A2–32b		2370	13.2095	37.8875	5	4	4	0	179.3	16.1	806.6	3.2	− 68.6	39.8	R
A2–33		2362	13.2095	37.8875	5	5	5	0	192.4	6.4	205.3	5.4	− 69.5	0.2	R
A2–34		2354	13.2098	37.8877	5	5	5	0	167.6	− 11.7	423.0	3.7	− 75.8	98.2	R
A2–35		2350	13.2099	37.8879											
A2–36		2340	13.2102	37.8885	6	6	6	0	176.8	− 17.1	268.2	4.1	− 84.5	73.3	R
A2–37		2326	13.2104	37.8877	5	5	5	0	175.0	− 16.4	708.9	2.9	− 83.1	83.8	R
A2–38		2310	13.2116	37.8873	5	5	5	0	179.5	− 24.6	348.6	4.1	− 89.4	94.9	R
A2–39		2295	13.2131	37.8861	6	6	6	0	188.4	16.4	405.6	3.3	− 66.9	16.3	R
A2–40		2280	13.2148	37.8851	4	4	3	1	47.6	− 33.9	173.9	7.0	33.3	161.0	N-i
A2–41		2248	13.2182	37.8838	5	4	4	0	355.2	− 8.1	137.7	7.9	72.1	233.6	N
A2–42a		2204			5										
A2–42a_up		2204	13.2265	37.8846		2	2	0	355.2	− 0.7	207.7	17.4			*n.a.*
A2–42a_lo		2203	13.2267	37.8848		3	3	0	338.5	− 3.9	233.8	8.1			*n.a.*
A2–42b		2194	13.2273	37.8861	6	5	5	0	341.3	− 14.9	102.1	7.6	62.1	260.7	N
A2–43		2184	13.2291	37.8868	5	5	5	0	344.3	− 13.9	132.4	6.7	64.4	256.3	N
A2–44		2145	13.2345	37.8909	5	5	5	0	354.5	− 39.3	196.4	5.5	54.1	226.6	N
A2–45		2140	13.2374	37.8917	5	5	3	2	2.7	− 47.5	57.7	10.2	48.1	214.3	N
A2–46		2113	13.2472	37.8878											
A2–47		2096	13.2519	37.8870											
A2–48		2081	13.2552	37.8891	5	5	5	0	11.0	1.7	249.9	4.8	73.5	175.7	N
A2–49		2003	13.2568	37.8871	6	6	6	0	347.5	37.3	409.6	3.3	75.9	342.0	N
A2–50		1976	13.2569	37.8846											

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Table 1. Continued

Flow	DG	Altitude (m)	Locality Latitude (° N)	Locality Longitude (° E)	n_0	n	n_l	n_{gc}	D (°)	I (°)	k	α_{95} (°)	VGP Latitude (° N)	VGP Longitude (° E)	Pol.
A2–51		1955	13.2566	37.8829	5	4	4	0	12.7	22.6	594.3	3.8	77.5	133.4	N
A2–52		1937	13.2583	37.8820	6	6	6	0	192.4	− 29.7	475.1	3.1	− 77.7	293.9	R
A2–53		1921	13.2591	37.8827	5	3	3	0	20.8	− 13.4	270.2	7.5			*n.a.*
A2–54		1917	13.2588	37.8829	6	6	5	1	15.3	− 24.6	389.3	3.4	59.8	187.2	N
A2–55		1905	13.2597	37.8826	6	6	6	0	11.9	− 16.3	155.9	5.4	65.4	188.5	N
A2–56		1890	13.2581	37.8814	6	6	6	0	205.0	− 38.0	191.4	4.9	− 64.8	285.7	R
A2–57		1880	13.2581	37.8814	5	5	5	0	208.8	− 35.5	68.9	9.3	− 61.7	291.0	R
A2–58		1870	13.2581	37.8814	6	6	6	0	210.0	− 43.9	121.9	6.1	− 59.2	279.5	R
A2–59		1864	13.2597	37.8809	6	6	6	0	357.6	− 11.9	214.9	4.6	70.6	225.1	N
A2–60		1860	13.2595	37.8810	6	6	6	0	357.9	− 17.5	43.2	10.3	67.7	223.4	N
A2–61a		1844	13.2616	37.8796	6	6	6	0	345.4	25.1	103.7	6.6	75.8	309.2	N
A2–61b		1827	13.2665	37.8817	7	6	6	0	1.6	0.3	357.1	3.6	76.8	210.9	N
A2–62		1822	13.2665	37.8817	6	6	6	0	356.2	8.3	415.3	3.3	80.2	240.6	N
A3–01		1798	13.2670	37.8810	6	6	4	2	187.0	− 6.0	153.6	5.4	− 77.6	3.3	R
A3–02		1781	13.2700	37.8797	6	6	6	0	180.6	− 15.6	117.2	6.2	− 84.6	31.5	R
A3–03		1738	13.2724	37.8785	5	5	5	0	177.7	3.8	214.9	5.2	− 74.7	46.6	R
A3–04		1724	13.2743	37.8787	6	6	6	0	177.4	− 30.8	137.1	5.7	− 85.8	181.1	R
A3–05		1700	13.2761	37.8794	6	6	6	0	191.5	10.3	118.8	6.2	− 68.3	5.4	R
A3–06		1695	13.2761	37.8789	6	6	6	0	171.5	− 26.4	212.5	4.6	− 81.7	133.5	R
A3–07					10										
A3–07a		1684	13.2764	37.8765		5	5	0	163.7	− 32.5	493.2	3.4	− 73.7	145.6	R
A3–07b		1681	13.2769	37.8764		4	4	0	208.4	41.3	224.6	6.1			*n.a.*
A3–08		1670	13.2792	37.8774	6	6	6	0	166.3	4.2	81.4	7.5	− 69.5	80.3	R
A3–09		1650	13.2798	37.8766	6	6	6	0	187.6	29.0	781.4	2.4	− 60.3	23.0	R
A3–10		1640	13.2789	37.8755	6	6	6	0	190.7	30.2	354.1	3.6	− 58.7	17.8	R
A3–11		1612	13.2775	37.8740	5	5	5	0	179.9	− 15.2	134.4	6.6	− 84.5	38.9	R
A3–12		1592	13.2793	37.8720	9	8	3	5	155.2	− 3.1	124.8	5.0	− 62.8	104.4	R
A3–13		1575	13.2791	37.8700	6	6	4	2	173.6	− 27.9	45.2	10.1	− 83.6	142.6	R
A3–14		1540	13.2792	37.8690	6	5	3	2	185.2	3.3	260.7	4.7	− 74.2	18.4	R
A3–15		1491	13.2805	37.8696	6	6	6	0	192.3	− 28.9	99.8	6.7	− 77.9	296.2	R
A3–16		1468	13.2802	37.8669	6	6	6	0	187.7	− 14.7	80.8	7.5	− 80.5	344.6	R
A3–17		1430	13.2820	37.8628	6	6	5	1	169.8	− 11.1	169.8	5.2	− 77.3	91.4	R
A3–18		1417	13.2826	37.8614	6	6	4	2	167.7	− 8.1	173.4	5.1	− 74.8	91.8	R
A3–19		1397	13.2837	37.8620	8	8	8	0	205.3	− 28.6	177.6	4.2	− 65.4	300.2	R
A3–20		1365	13.2848	37.8603	8	8	5	3	126.3	59.8	231.1	3.7	− 16.7	77.5	R-i
A3–21		1337	13.2899	37.8564	8	8	8	0	184.9	− 22.9	273.7	3.4	− 85.0	323.2	R
A3–22		1321	13.3031	37.8567	8	8	8	0	159.3	− 22.1	416.7	2.7	− 69.7	125.0	R
A3–23		1305			8										
A3–23_up		1305	13.3100	37.8542		3	2	1	352.4	− 33.4	246.0	7.9			*n.a.*
A3–23_lo		1304.5	13.3100	37.8542		4	4	0	165.4	5.4	861.1	3.1			*n.a.*
A3–24		1247	13.3231	37.8628	8	8	8	0	206.0	12.3	211.0	3.8	− 57.6	343.4	R
A3–25		1234	13.3408	37.8788	7	7	7	0	205.0	36.4	322.0	3.4	− 48.4	1.2	R
						Directional goups (DGs)									
A2–03 — A2–09	DG1					44			174.6	9.0	132.5	1.9	− 71.5	55.1	R
A2–20 — A2–21	DG2					12			215.0	40.7	466.7	2.0	− 40.0	354.5	R-i
A2–23 — A2–25	DG3					15			176.7	− 37.8	241.9	2.5	− 81.4	196.9	R
A2–26 — A2–29	DG4					21			189.8	18.0	253.7	2.0	− 65.6	13.9	R
A2–36 — A2–37	DG5					11			176.0	− 16.8	387.7	2.3	− 83.9	78.7	R
A2–42b — A2–43	DG6					10			342.8	− 14.4	123.2	4.4	63.3	258.7	N
A2–44 — A2–45	DG7					10			358.3	− 43.4	70.0	5.8	51.5	220.4	N
A2–56 — A2–58	DG8					17			207.8	− 39.4	97.3	3.6	− 62.1	285.1	R
A2–59 — A2–60	DG9					12			357.7	− 14.7	71.9	5.2	69.1	224.3	N
A3–09 — A3–10	DG10					12			189.2	29.7	447.6	2.1	− 59.4	20.3	R
A3–17 — A3–18	DG11					12			168.8	− 9.6	170.5	3.3	− 76.1	91.5	R

S_W is then approximated by $S_W = 81/K^{1/2}$. Following these calculations using the data set given in Table 1, our determined stable-polarity 79 VGPs for LL section allow us to obtain S_F of 17.0°, with lower and upper 95 per cent confidence level of −1.7°/+2.1° (Cox 1969) (Table 2). Additionally, even when thoroughly considered a data set of site-mean VGPs, in which specimen-level data from a single core are averaged and subsequently generated core-level data for each site are used in the calculation of the site-mean statistics, the resulting S_F value become nearly same as the former estimate.

For the purpose of robust estimation of palaeomagnetic pole and PSV, some studies documented that successive flows with similar palaeomagnetic directions in a statistical sense need to be grouped into the equivalent of a single cooling unit, so an independent direction is obtained (e.g. Mankinen *et al.* 1985; Chenet *et al.* 2008). In cases with high-density sampling, back-to-back flows often show little difference in their palaeomagnetic directions, presumably because they often erupted in a relatively rapid period. We use the simple method described by Mankinen *et al.* (1985) with the statistics test of McFadden & Lowes (1981) to detect a difference between neighbouring directions and subsequently the directions determined as 'same' are given a 'directional group (DG)'. That is, if the mean directions of two or more flows exhibit an overlap of their 95 per cent confidence ovals but no systematic trends in directional change, specimen-level directions from their flows are

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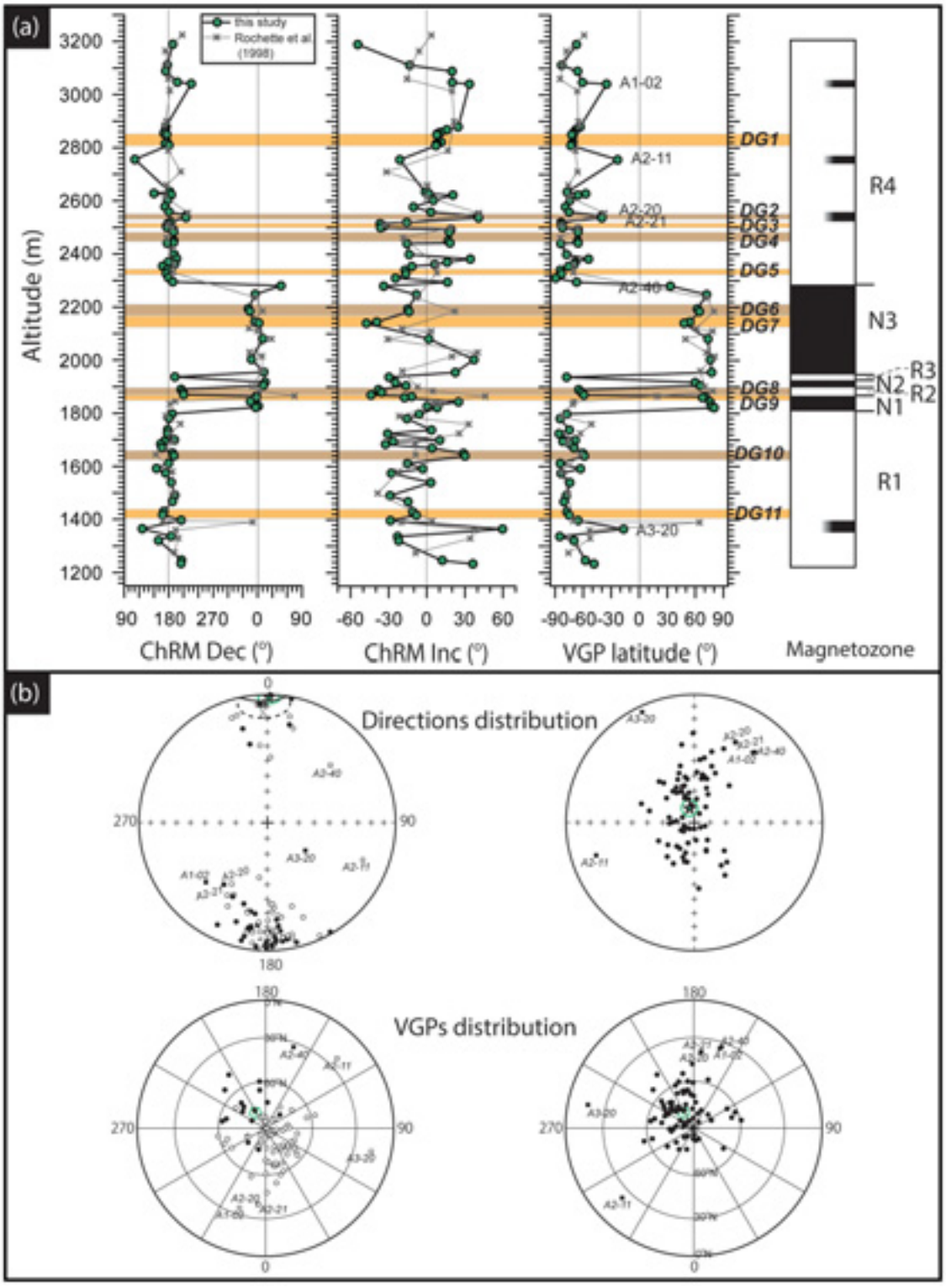


Figure 4. Summary of palaeomagnetic data (totally 85) for the LL section from this study. (a) Altitudinal variation of palaeomagnetic data (declinations and inclinations of flow-mean ChRMs, and the corresponding VGP latitudes; green circles), with the previous data by Rochette *et al.* (1998, grey x marks), and the resulting magnetostratigraphy. The defined DGs (see Section 4.4 for details) are given as color-shaded boxes. (b) Stereographic projections of the directions (the two plots in the upper row) and the VGPs (the two plots in the lower row). Solid (open) symbol in the direction projections shows downward (upward) inclination, and in the VGP projections they shows projection onto the northern (southern) hemisphere. The second projection of the directions is centred on the expected direction from the Africa reference pole of Besse & Courtillot (2002, 2003), $D = 3.8^\circ$ and $I = 11.7^\circ$. In the fourth projection of the VGPs, the original data with reversed polarity are converted into the equivalent data with normal polarity. Square symbols with letters of flow name indicate intermediate polarity data which are determined following the method of Vandamme (1994, see the text for details). Star symbol with the surrounding oval in green indicates the mean with a radius equal to α_{95} or A_{95} confidence limit from the directions or the VGPs.

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Table 2. ASD estimates and mean VGP positions obtained from our VGP data of LL section. N, number of VGPs used in calculation; Pol. magnetic polarities which the VGPs used present (N: normal, R: reversal, I: excursional/transitional); Cut-off, optimum cut-off angle calculated from the Vandamme (1994) method; S_P (S_F), total ASD (between-site ASD corrected for the within-site dispersion); $S_{F\text{-up}}$ ($S_{F\text{-lo}}$), upper (lower) limit of the S_F value by Cox (1969); pLat (pLon), latitude (longitude) of the calculated mean VGP; K, Fisherian precision parameter of the mean VGP; A_{95}, 95 per cent confidence limit on the mean VGP. See Section 4.4 for details.

Description	N	Pol.	Cut-off (°)	S_P (°)	S_F (°)	$S_{F\text{-up}}$ (°)	$S_{F\text{-lo}}$ (°)	pLat (°)	pLon (°)	K	A_{95} (°)
Using flow means											
Without cut-off	85	N + R + I		21.8	21.7	24.3	19.6	76.7	210.9	14.4	4.2
With cut-off	79	N + R	35.9	17.1	17.0	19.1	15.3	78.2	213.0	22.7	3.4
Using DG means instead of part of the flow means											
Without cut-off	65	N + R + I		22.7	22.6	25.7	20.1	77.4	211.6	13.6	5.0
With cut-off	60	N + R	36.6	17.6	17.5	20.0	15.5	78.9	212.4	23.1	3.9

averaged together, and the resulting mean direction are given a DG number. This procedure is made iteratively for all pairs of two or more neighbouring flows. It is considered that the flows in the same DG number are emplaced separately but, as a whole, in a very short time (not sufficient to record PSV). This allows to make 11 DGs, and their individual mean directions are given in Table 1, providing 65 independent mean directions for a re-examination of the afore-determined pole and PSV estimate. From the 65 VGP data, the Vandamme (1994) method allows excluding five means, that is, A1–02, A2–11, A2–21, A2–40 and A3–20 (whose polarities are thus considered excursional or transitional), in the calculations. The resultant overall mean VGP and S_F estimate is, thus, Lat. = 78.9° N and Lon. = 212.4° E (A_{95} = 3.9°), and 16.9° (15.0–19.3° in 95 per cent confidence interval), respectively (Table 2). It is good to note that both of the VGP and ASD estimates are statistically identical between the cases where the 'DGs' are defined and not in the calculations. For this reason, we decide to use a set of our flow-mean data without 'directional grouping' in the following discussion on the Oligocene geomagnetic field behaviour (Section 5.3).

5 DISCUSSION

5.1 Correlating the new LL magnetostratigraphy with the latest GPTS

Prior to considering the correlation of the new LL magnetostratigraphy with GPTS2016 by using available radiometric ages, extraction of radiometric ages to be used shall be needed because, owing to few degrees of tilting of lava flows, correlation of stratigraphic levels between distant flows in this study area and beyond becomes ambiguous especially when only information of altitude without palaeomagnetic data is available. Accordingly, 17 radiometric ages are extracted: eight $^{40}Ar/^{39}Ar$, and three K-Ar ages for magnetozone R4, four $^{40}Ar/^{39}Ar$ and one K-Ar ages for magnetozone N3, and 1 $^{40}Ar/^{39}Ar$ age for magnetozone R1 (Table S1, Supporting Information). Addition to this, to compare theses ages to GPTS2016, the ages need to be recalibrated to reflect the standards used in GPTS2016, that is, the FCs standard from Kuiper *et al.* (2008), and the ^{40}K decay constant updated by Min *et al.* (2000). The recalibrated ages are also given in Fig. 5a and Table S1 in the Supporting Information.

Most of the ages for the R4 magnetozone overlap with reverse polarity chron C11r or C11n.1r, some of the ages for the N3 magnetozone overlap with normal polarity chron C11n.2n, and the age for the R1 magnetozone falls within reverse polarity chron C12n in GPTS2016. Unfortunately, using the refined magnetostratigraphy and recalibrated radiometric ages do not still make the correlation with GPTS2016 clearer. This emphasizes that new dating for the section and possibly a new age calibration of GPTS may be needed to make it more straightforward in future works. Nevertheless, we attempt to find a new constraint about the correlation as follows. Given the statistically overlapping ages between the lower and upper parts of magnetozone R4, it is reasonable to interpret that there was no significant temporal hiatus between flood basalt and shield volcano sequences. No significant pause in volcanism between N3 and R4 is also indicated by the overlapping ages between them and the recording of a transitional direction between them (flow A2–40). Since there are only two stable, normal polarity chrons between C11n.1r and C12r in the GPTS2016, at least one magnetozone in the LL magnetostratigraphy should correspond to a little-known, very brief polarity chron (i.e. Cryptochron) or geomagnetic excursion. The N1-to-R3 sequence appears not to correspond to the recording of a full series of geomagnetic reversal process as suggested by Valet *et al.* (2012), because of the absence of 'transitional' polarity recording around reversal boundaries. We thus envisage three possible frameworks in magnetostratigraphic correlation: [A] R1 (R4) corresponding to C12r (C11r), [B] R1 (R4) corresponding to C12r (C11n.1r) and [C] R1 (R4) corresponding to C11r (C11n.1r) (Fig. 5b).

Possibility [A] requires two very brief chrons in C12r or C12n or both. Accordingly, one might be possible to correlate N1 and N2 to Cryptochron C12r-2 and Cryptochron C12r-1, respectively. On the one hand, the possibility [B] has a requirement of 1 out of 3 normal polarity magnetozones to be associated with a very brief chron or excursion. Accordingly, one possible correlation is C12r (R1)—C12n (N1)—C11r (R2-to-R3)—C11n.2n (N3)—C11n.1r (R4), by correlating N2 to Cryptochron C11r-1. In both cases, the more important thing is the lowermost section representing C12r. Both of these two possible correlations certainly differ from the previous interpretation (Hofmann *et al.* 1997; Rochette *et al.* 1998). This notion is also in disagreement with the standard interpretation by Coulié *et al.* (2003) and Riisager *et al.* (2005) which stated the earliest flood basalt eruptions could not be older than C12n based on their radiometric dating from the basal basaltic units of OEFB outside our sampled area. Caution should be exercised when correlating R1 to C12r as it depends only on a single age data (the recalibrated age of E199, 31.5 ± 0.6 Ma; Table S1, Supporting Information). On the other hand, using common recalibrations for the FCs standard (Kuiper *et al.* 2008), for example, the age for the oldest basalt representing reverse polarity presented in Riisager *et al.* (2005) is firmly within C12n of GPTS2016. To the current evidence, it is hard to make new alternatives such as possibilities [A] and [B] firmer.

Possibility [C] is similar to the preferred interpretation of Rochette *et al.* (1998) but requires two very brief polarity chron(s)

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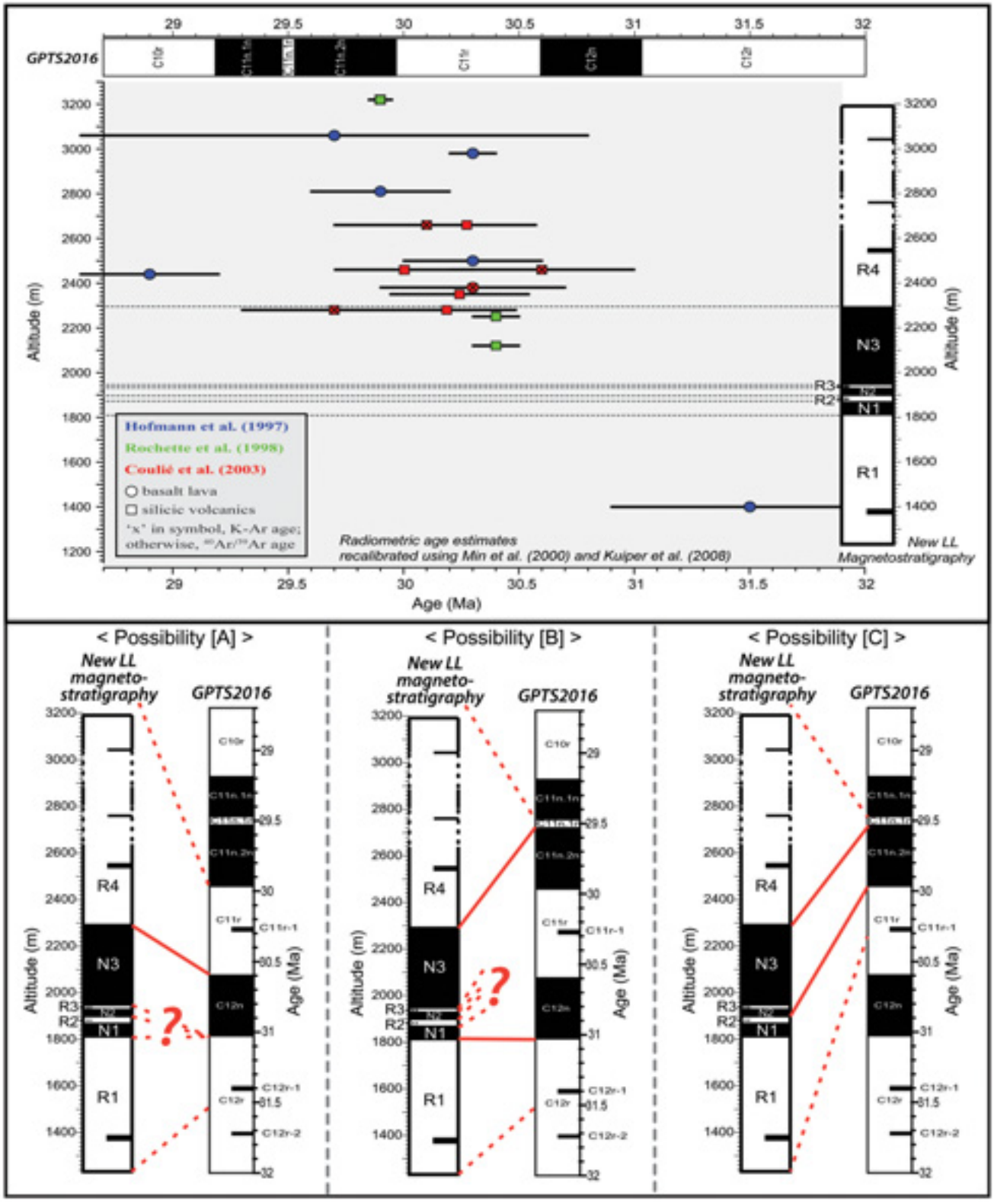


Figure 5. Summary of our magnetostratigraphic interpretations on correlating our LL magnetostratigraphy to GPTS2016 (Ogg *et al.* 2016) by using currently available radiometric ages. In the upper panel, recalibrated K-Ar and $^{40}Ar/^{39}Ar$ ages (Table S1, Supporting Information) with altitude are displayed, together with the LL magnetozone sequence of this study (on the right hand) and GPTS2016 (on the top). The originally reported radiometric ages of Hofmann *et al.* (1997), Rochette *et al.* (1998) and Coulié *et al.* (2003) for the LL section were recalibrated to the FCs standard with an age of 28.201 Ma by Kuiper *et al.* (2008) and the ^{40}K decay constant of Min *et al.* (2000), in order to use them in connecting between our LL magnetostratigraphy and GPTS2016. Symbols presented in the upper panel are the same as in Fig. 2(a). In the lower panel, three possibilities of correlation between the LL magnetostratigraphy and GPTS2016, which are considered in this study, are illustrated. See Section 5.1 for details.

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and/or excursion(s). Guidry *et al.* (2013) reported two very short polarity intervals from Equatorial Pacific deep-sea sediments (IODP site U1333), correlating them to the lowermost part of C11n.2n and the uppermost part of C11r (different from C11r-1) and thus interpreting as the record of putative excursions. The two putative excursions separated by a short period are consistent with R3 and N1 of the LL section. Accordingly, the two intervals of N2-to-N3 and R1-to-R2 might be correlated with C11n.2n and C11r in GPTS2016, respectively. This scenario may represent a refinement of the earlier interpretations (Hofmann *et al.* 1997; Rochette *et al.* 1998; Coulié *et al.* 2003; Riisager *et al.* 2005) without significant discrepancy from them (Fig. 5b).

In the possibility [C] framework (see Fig. 6), the low VGP latitude recorded by flow A3–20 (at ∼1365–1400 m altitude) might be associated with another excursion within C11r. A candidate corresponding to the excursional feature is the Cryptochron C11r-1, which was dated at 30.278–30.292 Ma by CK95, and calibrated to 30.263–30.286 Ma for GPTS2016, by using the ages of chron boundaries for CK95 as tie points and a simple linear interpolation between tie-point ages (C11n.2n/C11r, C11r/C12n–29.970 Ma, 30.591 Ma for GPTS2016; 30.098 Ma, 30.479 Ma for CK95). For magnetozone R1, there are three tie-points in this age–altitude model, i.e. 30.011 Ma–1820 m (N1/R1 boundary), 30.263 Ma–1397 m (top of the excursional feature), and 30.286 Ma–1365 m (base of the excursional feature). It leads to 1.68 and 1.39 m kyr^{-1} in an apparent average rate of the emplacement for the respective intervals between neighbouring two tie points. With an assumption of extrapolation by 1.39 m kyr^{-1} in emplacement rate, the age of the basal basalts at 1234 m is assigned as 30.38 Ma (alternatively, 30.37 Ma for 1.39 m kyr^{-1} in the rate). Similarly, for other magnetozones of the section it is possible to estimate the emplacement rates: ∼23 m kyr^{-1} for N1, the order of 1 m kyr^{-1} for the magnetozones between R2 and R3, < 0.8 m kyr^{-1} for N3 and > ∼18 m kyr^{-1} for R4; when assuming 600 000 km^2 as a constant areal extent, ∼14 km^3 yr^{-1} for N1, ∼0.4–4 km^3 yr^{-1} for R2-to-R3, and > 11 km^3 yr^{-1} for R4 (Fig. 6). These estimates possibly infer two pulses of accelerations in the volcanic activity. The inferred higher emplacement rates are roughly consistent with the very thick lavas in field observation.

To summarize, currently, a plausible magnetostratigraphic correlation for the LL section can be proposed as follows: C11r (R1-to-R2)—C11n.2n (N2-to-N3)—C11n.1r (R4), with N1 and R3 as the putative excursions in C11r and C11n.2n (Guidry *et al.* 2013) and the lowermost excursional feature representing C11r-1 (Fig. 6). However, it is emphasized that, to obtain a unique, firm solution in magnetostratigraphic correlation, new radiometric age constraints need to be added within the LL section and the GPTS ages need to be revised in this period as well.

5.2 Timing of the flood basalt and silicic volcanism: coincide with the Oi-2 event?

Our preferred magnetostratigraphic interpretation for the LL section (associated with the possibility [C] in Section 5.1; Fig. 6) allows us to discuss on the age of timing and duration of the OEFB-related flood basalt and silicic volcanism, and further their relative timing with respect to the Oi-2 cooling. It is possible that the flood basalt volcanism began just before C11r-1 (30.263–30.286 Ma, an inferred age interval for GPTS2016) or at ∼30.4 (more probably 30.38) Ma and terminated within C11n.1r (29.477–29.527 Ma for GPTS2016) or at ∼29.5 Ma, with the emplacement duration of ∼900 kyr and the apparent eruption rates of usually the order of 0.1–1 km^3 yr^{-1} but > 10 km^3 yr^{-1} at eruption peaks, and the silicic volcanism began within an early stage of C11n.2n (29.527–29.970 Ma for GPTS2016) or at ∼29.9 Ma and afterwards continued episodically. The reported age of the Oi-2 initiation in related literature is ∼30.3 Ma (Pekar & Miller 1996) or 30.35 Ma (Wade & Pälike 2004), which were estimated in the timescale of CK95. These ages can be relocated to 30.30 and 30.38 Ma, respectively, in GPTS2016, by a simple linear relationship between ages of C112n.2n/C11r and C11r/C12n boundaries in their own timescales. Given these Oi-2 ages and our magnetostratigraphic constraints on the timing of the flood basalt and silicic volcanism, it is inferred that the Oi-2 event postdates the earliest flood basalt volcanism by ∼100 kyr or less and predate the earliest voluminous silicic volcanism by ∼400–480 kyr.

Alternatively, if the recalibrated radiometric ages (Fig. 5a and Table S1, Supporting Information) alone are taken into account, the relative timing of the Oi-2 event with the eruptions becomes different significantly from the former interpretation. The Oi2 event is inferred to postdate the earliest flood basalt by 1120 kyr or more, given the age of 31.5 Ma in the lower R1 for the LL section. Then, it is inferred to postdate the earliest silicic eruption by 20 kyr or more, based on the oldest age of 30.4 Ma in the upper N3 for the section.

Different ways of causal relationships between volcanism and global climate impact over geological timescales have been suggested. According to Kasbohm & Schoene (2018), voluminous flood basalts are thought to drive global climate optimum over short geological timescales like in the Columbia River flood basalt (CRFB) and the mid-Miocene climate optimum (MMCO). These authors have refined the chronology and duration of CRFB volcanism and correlated the onset of the CRFB volcanism and the MMCO beginning to within ∼100 kyr. Their age model also indicated that the MMCO continued for > 1 Myr after the termination of the majority of CRFB volcanism. Alternatively, flood basalts are even thought to be capable of triggering global cooling impact or glaciation. Lefebvre *et al.* (2010) demonstrated, by using a coupled biogeochemical energy balance model, that chemical weathering of voluminous basalt province can lower CO_2 thereby leading to a long-term cooling or glaciation for several millions of years even after the culmination of volcanism. Their model also explained possible warming during and just after the volcanism caused by the atmospheric CO_2 increase. On the other hand, voluminous silicic volcanism is also thought to drive short-term global cooling by sulfur gas emission which results in preventing solar radiation from reaching the Earth surface (e.g. Mussard *et al.* 2014). According to this driving mechanism, the gas emission during volcanism possibly causes climate cooling contemporaneously in geological time.

Currently, it is difficult to validate a causal link between OEFB and Oi-2 events. The above-interpreted, two different constraints on the relative timing between them bring about contrasting inferences about the possibility of their causal linkage. Discussion on these is detailed below.

Considering the constraints of their timing based on the magnetostratigraphic correlation, first, the earliest silicic volcanism at ∼29.9 Ma cannot drive the Oi-2 cooling at 30.3 or 30.38 Ma. On the other hand, the flood basalt volcanism may have a potential to play a causal role in the Oi-2 cooling, based on the ∼900 kyr duration, the eruption rates of > 10 km^3 yr^{-1} at peaks, and the

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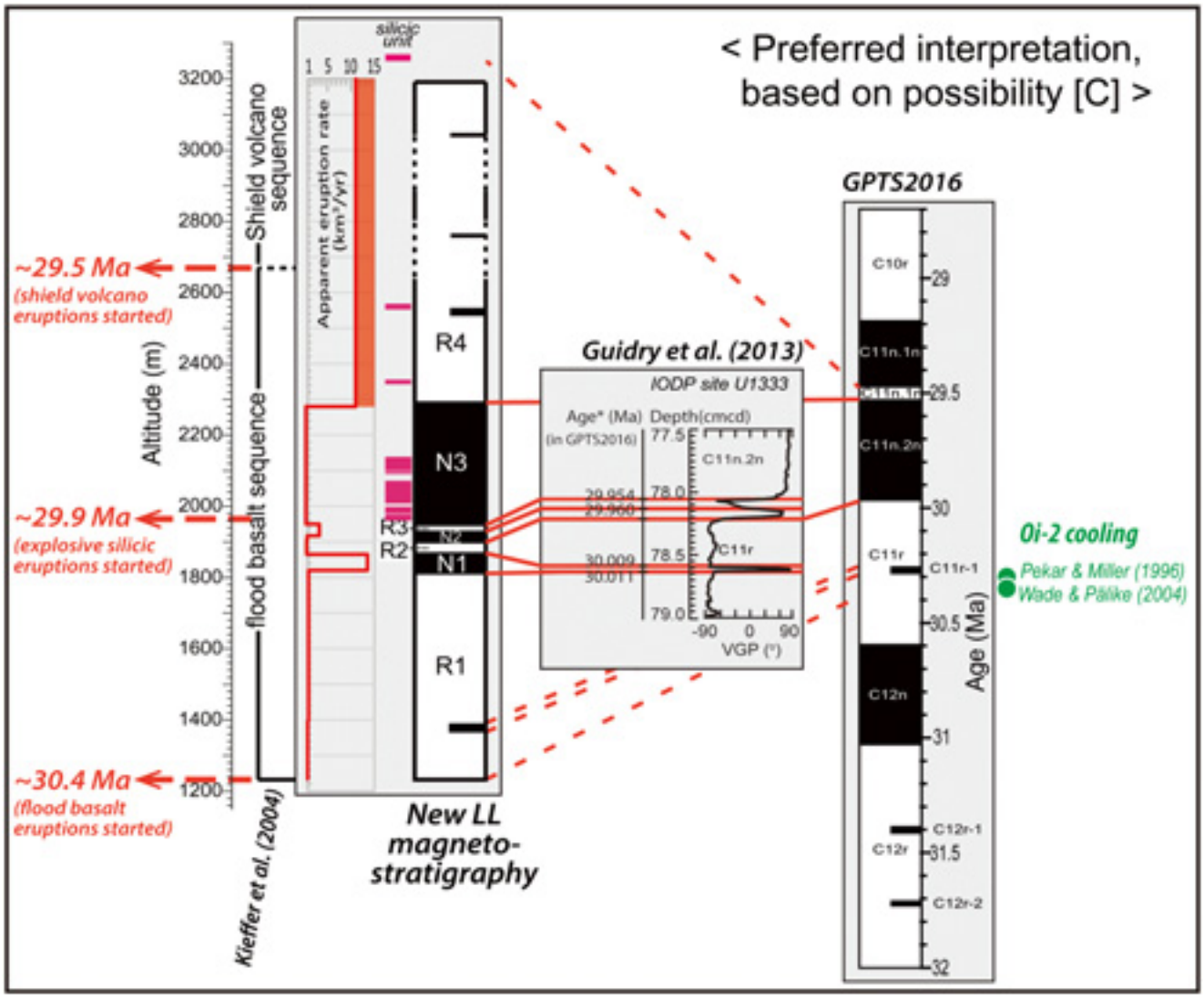


Figure 6. An illustration of our preferred interpretation on correlation between our LL magnetostratigraphy and GPTS2016 and the timing of the OEFB flood basalt and silicic volcanism, and their relative timing with the Oi-2 cooling. The division of basaltic sequences in LL section shown is from Kieffer *et al.* (2004). The apparent eruption rates (in $km^3\ yr^{-1}$) are estimated, with assumption of 600 000 km^2 as a constant areal extent, by using tie-point ages that were made by correlation with the timescales of Guidry *et al.* (2013) and GPTS2016 and then re-calculated to be those of the GPTS2016 framework; red line with shaded box denotes apparent eruption rate estimate of no less than the red line value. The age estimates at the base and top of the putative excursions in C11r and C11n.2n, respectively (Guidry *et al.* 2013), are relocated based on the age model of GPTS2016 in this study. The ages for Cryptochrons C11r-1, C12r-1 and C12r-2 presented in GPTS2016 are tentative estimates presented in this study. Note that the shown Oi-2-cooling ages of ∼30.3 and 30.35 Ma in original studies (Pekar & Miller 1996; Wade & Pälike 2004), which adopted the CK95 timescale, can be relocated to ∼30.30 and 30.38 Ma, respectively, in GPTS2016. See Sections 5.1 and 5.2 for details.

temporal correlation with the Oi-2 event within < 100 kyr, similar as observed for other large igneous provinces possibly associated with global environmental impacts (e.g. Burgess & Bowring 2015; Schoene *et al.* 2015; Kasbohm & Schoene 2018). However, Lee *et al.* (2016) and Brune *et al.* (2017) proposed that enormous and prolonged CO_2 release from continental rifting could cause contemporary global warming. Therefore, given the OEFB volcanism accompanied by the Afro-Arabian plate breakup, it is likely reasonable that during the OEFB volcanism climate warming should have been prevailingly prolonged. In contrast, considering the timing constraints based mainly on the recalibrated radiometric ages for the LL section, both flood basalt eruption starting at > ∼31.5 Ma and silicic eruption starting at > ∼30.4 Ma may be possible to trigger the Oi-2 cooling (which occurred at 30.3 or 30.8 Ma) via the above-described different mechanisms. Therefore, the low-resolution of the available radiometric ages coupled with their discrepancy from the GPTS2016 age model hinder resolving the issues of the relative timing of the OEFB eruptions, the Oi-2 cooling, and their causal linkage. Further efforts in providing high-resolution radiometric ages for the LL section and revising the GPTS ages for the Early Oligocene can potentially provide us with a clearer view on these.

5.3 Implications for the Early Oligocene field behaviour

Fig. 7(a) shows a projection of our overall mean VGP of the LL section, together with previously reported coeval poles from same and surrounding regions (Rochette *et al.* 1998; Ubangoh *et al.* 1998;

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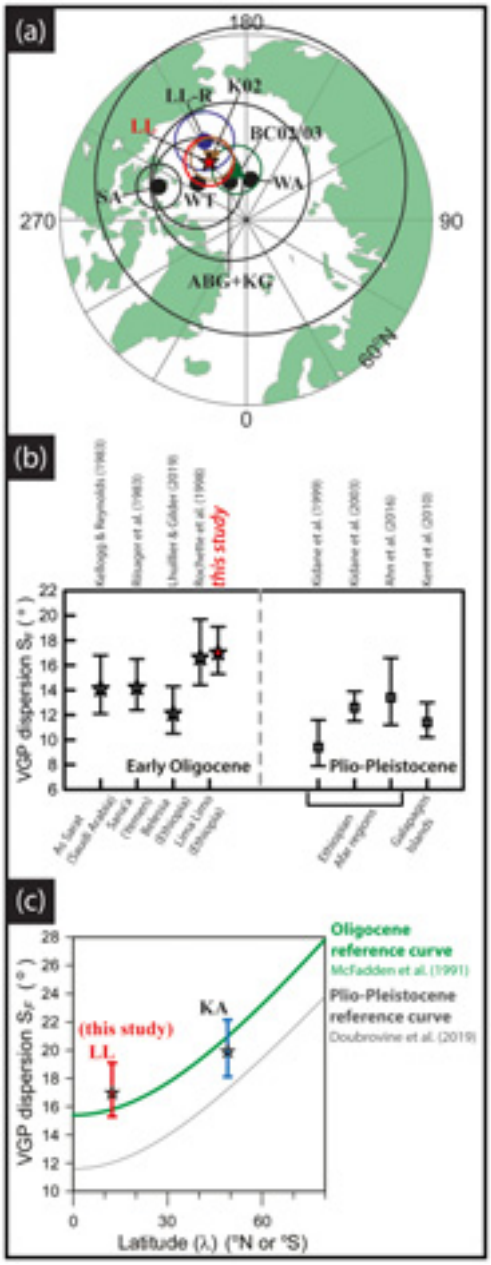


Figure 7. Comparison of palaeomagnetic poles and VGP dispersions (S_F) for the Oligocene. (a) Stereographic projection of palaeomagnetic poles with A_{95} confidence limit (radius centred on the pole) during the Oligocene: LL (red star), Lima Limo (Ethiopia) of this study; LL-R, Lima Limo of Rochette *et al.* (1998); WT, Wegel Tena (Ethiopia) of Rochette *et al.* (1998); ABG + KG, Abbay Gorge and Kessem Gore (Ethiopia) of Kidane *et al.* (2002); WA, Cameroon, west Africa of Ubangoh *et al.* (1998); SA, Sana'a (Yemen) of Riisager *et al.* (2005). 30 Ma synthetic reference poles of Africa are also shown: BC02/03, Besse & Courtillot (2002, 2003); K02, Kidane *et al.* (2002). (b) Comparison of the VGP dispersions (S_F) at low latitudes for the (Early) Oligocene and the Plio-Pleistocene. See Section 5.3 for details. (c) VGP dispersion (S_F) versus latitude plot of two robust Oligocene data at different latitudes, that is, the LL 13° N) data ($N = 79$) of this study and the Kerguelen Archipelago (KA, 49° S) data ($N = 94$) of Plenier *et al.* (2002). Reference Model G fitting curves for the Plio-Pleistocene (Doubrovine *et al.* 2019 who present the 0–5 Ma Model G curve fitting to the data of Opdyke *et al.* 2015) and for the Oligocene (McFadden *et al.* 1991; to be more exact, the 22.5–45 Ma time window) are also shown for discussion (see Section 5.3).

Kidane *et al.* 2002; Riisager *et al.* 2005) and the synthetic African poles for ~30 Ma by Besse & Courillot (2002, 2003) and Kidane *et al.* (2002). Our LL pole position is close to that calculated from the Rochette *et al.* (1998) data when excluding excursional/transitional and poorly defined flow-mean data (75.7° N and 207.9° E, with $N = 37$ and $A_{95} = 4.7°$). Still, it is evident that our pole lies further northward than the previous pole and becomes more consistent with the other coeval poles. It is thus concluded that even for the LL section such strong non-dipole component suggested by Rochette *et al.* (1998) is negated, advocating the argument of Riisager *et al.* (2005).

Fig. 7(b) shows the new S_F estimate for the LL section, together with other Early Oligocene and Plio-Pleistocene S_F values from low site latitudes. Our LL S_F, 17.0° (15.3−19.1° in 95 per cent confidence interval), is nearly identical with the 14.4−19.7° interval recalculated by Lhuillier & Gilder (2019) for the Rochette *et al.* (1998) data. On the other hand, Lhuillier & Gilder (2019) yielded an S_F of 14.2° (13.2−15.4° in 95 per cent confidence interval) based on a VGP data set ($N = 167$) compiling their Belessa section data and the previous data from LL (Rochette *et al.* 1998), Sana'a (Riisager *et al.* 2005) and As Sarat (Kellogg & Reynolds 1983). This estimate is somewhat lower than our LL S_F. Although currently no plausible explanation for the slight difference can be given, the Belessa section might involve a younger volcanic sequence in the proximity of Belessa (Kieffer *et al.* 2004). The two younger K-Ar ages obtained by Lhuillier & Gilder (2019), which were sidestepped in their correlations, and the multiple sizable palaeosols along their section are consistent with this interpretation. The validity of our S_F estimate as PSV is supported by the use of data set consisting of recordings at entirely different times with the ~0.9 Myr (or longer) duration as well as the amount of data. Given the reported Plio-Pleistocene S_F estimates for low site latitudes [7.9–11.6°, recalculated by Lhuillier & Gilder (2019) using the data in Kidane *et al.* (1999); 11.5–13.9°, calculated by Kent *et al.* (2010) using the data in Kidane *et al.* (2003); and 11.2–16.6°, calculated by Lhuillier & Gilder (2019) using the data in Ahn *et al.* (2016)], our data confirm that the VGP dispersion during the Early Oligocene was distinctly larger than that during the Plio-Pleistocene, possibly indicating dissimilarity in the time-averaged field geometry of the two periods.

In order to find an additional characteristic on the Oligocene field behaviour, we attempt to analyse a site (palaeo)latitude dependence of S_F, based on the Model G (McFadden *et al.* 1991), by combining our LL S_F estimate with the results from others at different site latitudes, and to compare it with those for the Plio-Pleistocene published in other literature (see Fig. 7c). An S_F variation curve as a function of the latitude of the Model G is expressed with the shape parameters a and b following the eq. (4) of McFadden *et al.* (1991): $S_F(\lambda) = \{a^2 + (b\lambda)^2\}^{1/2}$, where λ is (palaeo)latitude of the sampling site. The b/a ratio for Model G represents a degree of latitude dependence of S_F; the higher b/a ratio correspond to the stronger latitude dependence of S_F. McFadden *et al.* (1991) initially provided estimates of the Model G shape parameters for the 22.5–45 Ma period including the Oligocene, that is, a ~15.4 and b ~0.29 (b/a ~0.019). However, in their 22.5–45 Ma Model G curve the latitude-binned S_F data at latitudes of $< 40°$ does not fit well with the curve. Besides this, after their pioneering work, Biggin *et al.* (2008) pointed to using a constant value for the S_W correction that is suspected to bias their S_F estimates resulting in underestimated S_F values at low latitudes and vice versa. We first attempt to confirm the validity of the original shape parameters for the Oligocene by using robust S_F data, which are recently published after the McFadden *et al.*'s contribution with a large number of palaeomagnetic data

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in a single study area. Plenier *et al.* (2002) estimated S_F of 19.95° from 94 VGP data from Oligocene lava sequences of the Kerguelen Archipelago (∼49° S; labelled 'KA' in Fig. 6c). The LL and KA S_F estimates, when their 95 per cent confidence intervals are considered, overlap with the Oligocene Model G curve of McFadden *et al.* (1991), but their mean values are slightly higher at the low latitude and slightly lower at the high latitude, respectively, than the values predicted from the previous Model G curve (Fig. 7c). These two data allow a simple calculation of the shape parameters to a ∼16.8 and b ∼0.22, indicating slightly weaker latitudinal dependence of S_F (b/a ∼0.013). For the Model G curve of the Plio-Pliocene (or 0–5 Ma), Doubrovine *et al.* (2019) have recently updated the values of the shape parameters based on the data of Opdyke *et al.* (2015); a ∼11.6 and b ∼0.26, with b/a ∼0.022. Thus, our results support possibly a weaker latitudinal dependence of VGP dispersion (S_F) for the (Early) Oligocene than the Plio-Pleistocene.

6 CONCLUSIONS

Through high-density resampling of OEFB volcanic rocks along the ∼2 km tall LL section (Ethiopia), new, high-resolution palaeomagnetic results supported by rock magnetism were obtained and lead to the following conclusions:

(i) Collectively 85 well-defined flow-mean directions have been obtained passing the reversal test. The corresponding flow-mean VGPs are used in constructing magnetostratigraphy, which consists of seven magnetic polarity zones (R1−N1−R2−N2−R3−N3−R4 from bottom to top) being separated by six full polarity reversals, with six different transitional/excursional records.

(ii) Correlating new LL magnetostratigraphy to GPTS2016 with previous age estimates make no unique solution but strongly suggests the recording of at least one very brief magnetochron or geomagnetic excursion. We currently propose a preferred correlation: C11r (R1-to-R2)—C11n.2n (N2-to-N3)—C11n.1r (R4), involving N1 and R3 as putative excursions in C11r and C11n.2n, respectively (which were initially been reported by Guidry *et al.* 2013). This has broadly no conflict with the earlier interpretations but a clear improvement.

(iii) Despite the improvement in magnetostratigraphy, the low-resolution of the radiometric ages and their discrepancy from the age model of GPTS2016 hinder making clearer constraint on the relative timing of the OEFB eruptions, the Oi-2 cooling, and their causal linkages. New high-resolution radiometric ages for the LL section and revision of the age model of GPTS2016 for the Early Oligocene will be required to resolve this through future works.

(iv) Our overall mean pole for the LL section of this study, Lat. = 78.2° N, Lon. = 213.0° E (A_{95} = 3.4°, N = 79), is not far apart from other coeval palaeomagnetic poles from different regions and the reference pole of Africa, thereby solving the problem reported by Rochette *et al.* (1998).

(v) Our PSV estimate for the Early Oligocene, together with another, robust PSV estimate at a higher latitude, allow us to confirm that, compared to the geologically recent period (Plio-Pleistocene), the Early Oligocene field has higher degrees of PSV and a weaker latitudinal dependence of PSV.

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SUPPORTING INFORMATION

Supplementary data are available at *GJI* online.

Figure S1. Typical *Ms–T* curves from strong-field thermomagnetic analyses in vacuum. Heating (cooling) curve is displayed in black thick line (grey thin line). '$Tc_{_heating}$' denotes the determined temperature(s) of *Tc* in the heating curve. See the text for details of the experimental conditions.

Figure S2. Summary of results from the VSM measurements of 92 samples (one sample per flow). (a) The Mrs/Ms versus Bcr/Bc plot of 92 analysed samples as shown in square symbols. Single domain (SD)-multi-domain (MD) mixing lines and superparamagnetic domain (SP)-SD mixing lines (with SP grains of 10 and 15 nm, respectively) proposed by Dunlop (2002) are shown as dashed lines. All boundary values for the magnetic domain regions (SD, PSD and MD) are from Day *et al.* (1977). (b) Stepwise acquisition curves of IRM for five typical samples. (c)–(g) Magnetic hysteresis loops of five typical samples. Back (grey) curve shows the raw measurement data (the resulting data after paramagnetic slope correction). The positions of the corresponding parameter data on the Mrs/Ms versus Bcr/Bc diagram are indicated in Fig. S2(a).

Figure S3. Typical demagnetization behaviours and the associated thermomagnetic curves for some of flows belonging to magnetozones newly presented in this study: (a) R3, (b) N2, (c) R2 and (d) N1 magnetozones, respectively. Left-hand (middle) panels, AF (TH) demagnetization results showing in the intensity decay curve, orthogonal vector plot, and occasionally equal-area projection; right-hand panels, *Ms–T* curves. In the orthogonal vector and equal-area projection plots, green symbols represent the data points used in determining a PCA linear or GC (planar) fit as a ChRM component. Magenta circle and line in the equal-area projection plot (only for specimens A2-51-b1 and A2-521-1) represent a ChRM vector determined. The statistics of the determined ChRMs for specimens are found in Table S3.

Table S1. K-Ar and $^{40}Ar/^{39}Ar$ age estimates previously published for the LL section (Ethiopia). Sample ID, original sample name therein from the literature. Age with abbreviation † represents a weighted mean of multiple age estimates originally reported, by following Taylor (1982). Ages with abbreviation ∗ are not taken into account for discussion in this study (see text); Pol., geomagnetic polarity determined in the original literature; K-Ar (recalibrated), recalibrated K-Ar age estimate (in Ma) using the updated ^{40}K decay constant (Min *et al.* 2000), where its 1σ uncertainty is same as the original value; $^{40}Ar/^{39}Ar$ (recalibrated), recalibrated $^{40}Ar/^{39}Ar$ age estimate (in Ma) using the updated ^{40}K decay constant (Min *et al.* 2000) and FCs standard with an age of Kuiper *et al.* (2008), where its 1σ uncertainty is same as the original value; Std, standard originally used as the neutron fluence monitors for $^{40}Ar/^{39}Ar$ dating; Ref, reference of age data. Blank means 'no data' or 'no applicable'. Part of these data are also shown in Fig. 2.

Table S2. Summary of rock magnetic results. NRM0 (with σ_{NRM0}), initial NRM intensity before any treatments (with associated standard deviation). NRM_{MAX}, maximum value of NRM intensities for specimens within a flow. Initial χ, initial magnetic susceptibility. MDF (with σ_{MDF}), median destructive field of decay curve in NRM intensity during progressive AFD experiments (with associated standard deviation). Mrs, remanent saturation magnetization. Ms, saturation magnetization. Hc, coercivity. Hcr, coercivity of remanence. Mr/Ms, the ratio of Mrs to Ms. Hcr/Hc, the ratio of Hcr to Hc. Max (Min) in $T_{demag95\%}$, $Tc_{_predo}$, and $Tc_{_highest}$, maximal (minimal) temperatures in the midst of which demagnetization reaches 95 per cent of intial NRM intensity ($T_{demag95\%}$), the Curie temperatures of the phase carrying the largest portion of magnetization ($Tc_{_predo}$), and the highest Curie temperatures ($Tc_{_highest}$s), respectively. Blank denotes 'not analysed'. Part of these data are also shown in Figs S1 and S2.

Table S3. Interpreted linear or GC (planar) fits of characteristic remenance directions for individual specimens by PCA. Dg and Ig: interpreted characteristic direction for each specimen. MAD: maximum angular deviation for the PCA interpretation. Pts.: number of data points used in the interpretation. PCA type: L, linear fit; GC, great circle fit. Demag type: A, alternative field demagnetizations; L-A, alternating field demagnetizations after the low-temperature treatment and T, thermal demagnetizations.

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